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Design and modeling of tetrahedral soft-legged robot for multi-gait locomotion

Yuxuan Wang, Jiangbei Wang, Yanqiong Fei

***Abstract*—This paper presents a novel tetrahedral soft-legged robot for multi-gait locomotion, including steering, crawling and rolling along three different directions. The proposed robot has a compact structure with four active soft legs plugged together onto a rigid pivot. Each active soft leg is made of a novel pneumatic omnidirectional bending soft actuator. Locomotion is achieved by the sequential bending actuation of soft legs. Different inflation sequences are used to drive the soft legs to bend and realize multi-gait locomotion. Multiple gaits are analyzed with dynamic simulation methods. By discretizing the continuous soft leg into rigid links hinged end-to-end, a multibody dynamics model is built to simulate the robot dynamics and verify the feasibility of multi-gait locomotion. The crawling speed can reach $3.7\,\mathrm{mm\,s^{-1}}$, the steering speed can reach $1.1\,^\circ\mathrm{s^{-1}}$, and the rolling speed can reach $5.5\,^\circ\mathrm{s^{-1}}$ and $20.6\,\mathrm{mm\,s^{-1}}$. Experimental and simulation results show the capability of multi-gait locomotion of the proposed robot. The proposed robot has potential applications in search and rescue.**

***Index Terms*—Multi-gait locomotion, soft robot, omnidirectional bending.**

I. Introduction

SOFT robots are created by applying soft materials such as elastomers[1], fabrics[2], shape memory alloy[3] and electroactive polymers[4] to robots and actuating them to perform the motion of continuous infinite degree of freedom (DOF). Compared with conventional rigid robots, soft robots are more flexible, adaptable and safe, allowing them to interact with unstructured environments better and perform tasks in more dynamic manners[5]. According to the functions, soft robots can be classified into four major categories: locomotion, manipulation, assistance and sensing. Most of the locomotion soft robots are inspired by natural creatures such as octopus[6], starfish[7,8], jellyfish[9], fish[10,11], turtle[9], frog[12], snake[13,14], inchworm (or caterpillar or earthworm)[15–19], spider[20], cockroach[21] or other quadruped[22,23] and hexapod[24,25] animals. Their movement patterns generally include swimming, crawling, jumping and rolling. In addition, there are some special locomotion soft robots that are not bio-inspired, such as the tensegrity robots based on extension springs[26–28], the JSEL rolling ball[29], the soft rolling wheel[30–33], the flipping Fifobot[34] and some jumping robots based on explosion[35,36] or buckling[37].

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These movement patterns have their own advantages and disadvantages in fitness, speed, acceleration, energy economy and stability[38]. In fitness, different locomotion patterns are adapted to different environments, such as swimming in the water, crawling on land, jumping over obstacles and rolling on a slope. In speed and acceleration, there is no doubt that jumping is the best locomotion (up to $0.6\,\mathrm{m}$ of vertical displacement in $0.7\,\mathrm{s}$)[35], but its stability is poor because the jumping direction and displacement are difficult to be controlled. In addition, jumping is often intermittent because it takes time to accumulate energy. Except for the jumping locomotion, the rolling has faster velocity ($0.3\,\mathrm{m\,s^{-1}}$)[32] than the swimming ($0.15\,\mathrm{m\,s^{-1}}$)[11] and crawling ($0.05\,\mathrm{m\,s^{-1}}$)[14], but rolling stability is vulnerable to the robot's inertia and the ground's slope. The crawling locomotion is generally more stable but slower than the rolling locomotion.

Most of the rolling soft robots are designed into a closed-chain structure like a wheel. They need at least three soft actuators to construct the soft rolling robot. In previous studies, such robots usually use four[32], six[30], eight[31] or more[29] soft actuators to achieve rolling locomotion. Although more actuators can ensure more smooth movement, they also make robots bulky and inflexible. In addition, more actuators take longer actuation time in one locomotion cycle and reduce the average moving speed. There's another kind of rolling robot is the tensegrity robot such as T12-R[26] and SUPERball v2[28]. Tensegrity robots usually have large volume, large mass, complex structure and poor structural compactness.

Most of the crawling soft robots are bio-inspired by the inchworm movement[16,19] Crawling soft robots utilize bending or extending soft actuators to simulate the curling and stretching motion of inchworms and thus mimic the crawling locomotion of inchworms. Generally, crawling soft robots employ fewer soft actuators than rolling soft robots, and have great potential in portability and untethered navigation. However, crawling soft robots the slip effect which reduce the speed of crawling locomotion[18,19].

The steering locomotion is a mode as important as the linear locomotion for robots to navigate over the whole ground space. In recent research, the steering movement can be achieved by the differential drive method based on a parallel structure[39] or by the biaxial method based on a quadrilateral structure[15,34] or a spherical structure[29]. The differential drive method has a turning radius that limits the maneuverability of the robots. The biaxial method only supplies finite direction for robots to move along and cannot change the pose of robots.

To enhance the adaptability to the environment, some soft robots are designed with multi-gait locomotion. Hu et al[40].

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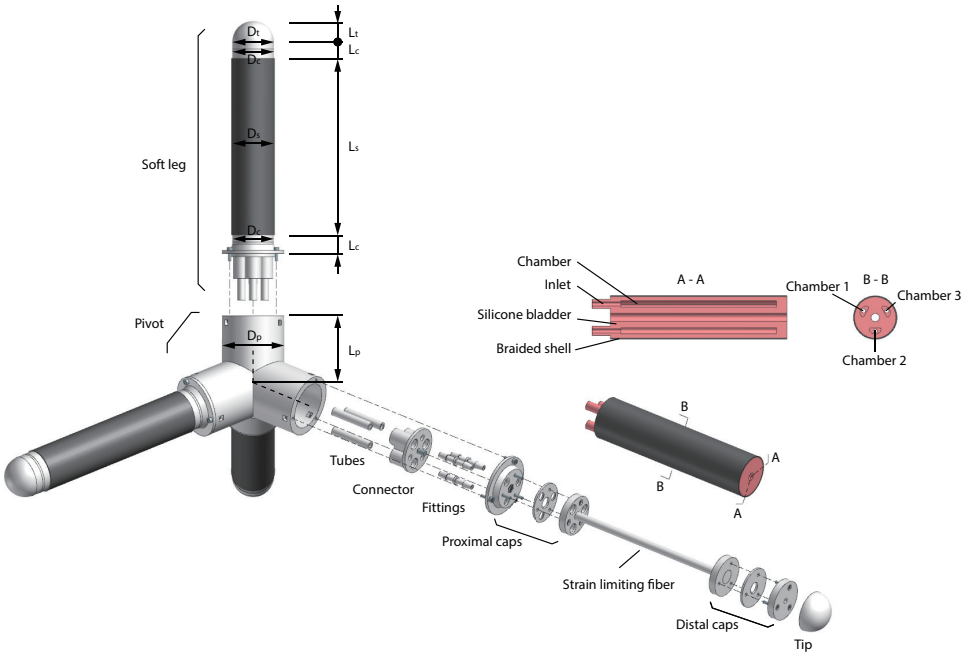


Fig. 1. **Structure of the tetrahedral soft-legged robot.** Four soft actuators of the robot are inserted into a 3D printed rigid body, and each soft leg can be bent in all directions to form a regular tetrahedron structure.

developed a type of magneto-elastic soft millimeter-scale robots that can swim inside and on the surface of liquids, climb liquid menisci, roll and walk on solid surfaces, jump over obstacles, and crawl within narrow tunnels. Ahn et al[41]. presented a light-powered soft robot that can crawl on the ground, squeeze its way through a small channel, and jump over a barrier. Du et al[31]. proposed a novel Shape Memory Alloy (SMA) actuated soft robot with rolling, Omega crawling and vermiculation locomotion. These soft robots have a compact strip-shaped structure integrated with soft actuators, and multiple locomotion modes are explored to extend the movement capability of soft robots. However, these soft robots are one-dimensional in morphology and only makes linear locomotion in theory.

Therefore, this research builds a tetrahedral soft-legged robot that bridges these four gaps: 1) in the rolling robots, the volume and weight are too large, which leads to the inconvenience of movement, and too many actuators will reduce the moving speed; 2) the crawling robots are affected by the slip effect so that their speeds are also slow; 3) the steering motion of robots can't achieve omni-directional turning and attitude adjustment at the same time; 4) most of the multi-gait robots can only realize linear motion.

We propose a tetrahedral soft-legged robot that can perform multi-gait locomotion including steering, crawling and rolling. The tetrahedral structure can reduce the volume and weight of the robot. And the structure allows the robot to stand and move on the ground by the least (i.e. three) legs stability, which avoids the inertia and slope vulnerability of the wheel-like or ball-like rolling soft robots. The proposed soft-legged robot has fewer (only four) soft actuators than the wheel-like or ball-like rolling soft robots. If the robot has fewer actuators, it will have smaller total volume and faster moving speed. This makes the robot have a shorter actuation sequence and faster rolling speed. We use extra soft actuators to change the frictional forces at the contact points with the ground, which can reduce the slip effect and increase the speed. The proposed soft-legged robot can perform the rolling and crawling locomotion along three different directions. In addition, the proposed soft-legged robot can change its pose and thus steer without changing position, which indicates better maneuverability than the differential drive and biaxial robots mentioned above.

II. Materials and Methods

A. Robot design

The proposed robot has a compact tetrahedral structure with four identical soft legs plugged onto a 3D-printed tetrahedral rigid pivot (Fig.1). Each soft leg is made of a pneumatic omni-directional bending soft actuator. A hollow cylindrical silicone bladder with three semicircle cross-section air chambers is fabricated by casting first. Then, the bladder is encircled by a braided shell in order to restrict the radial expansion of the bladder. Simultaneously, a flexible strain limiting fiber goes through the central hole of the bladder, and the proximal and distal caps on the two ends limit the extension of the bladder. The inlets of chambers go through the proximal caps and connect with the fittings. The fittings go through the connector and connect with the tubes. Air is pumped into or released from the chambers of the bladder through the tubes. Inflation of each chamber causes the soft actuator to bend towards the opposite side of the chamber, and three chambers are sufficient and necessary to achieve omnidirectional bending of the soft actuator. The semisphere-shaped toe is made of silicone to improve the contact condition between the robot and the ground.

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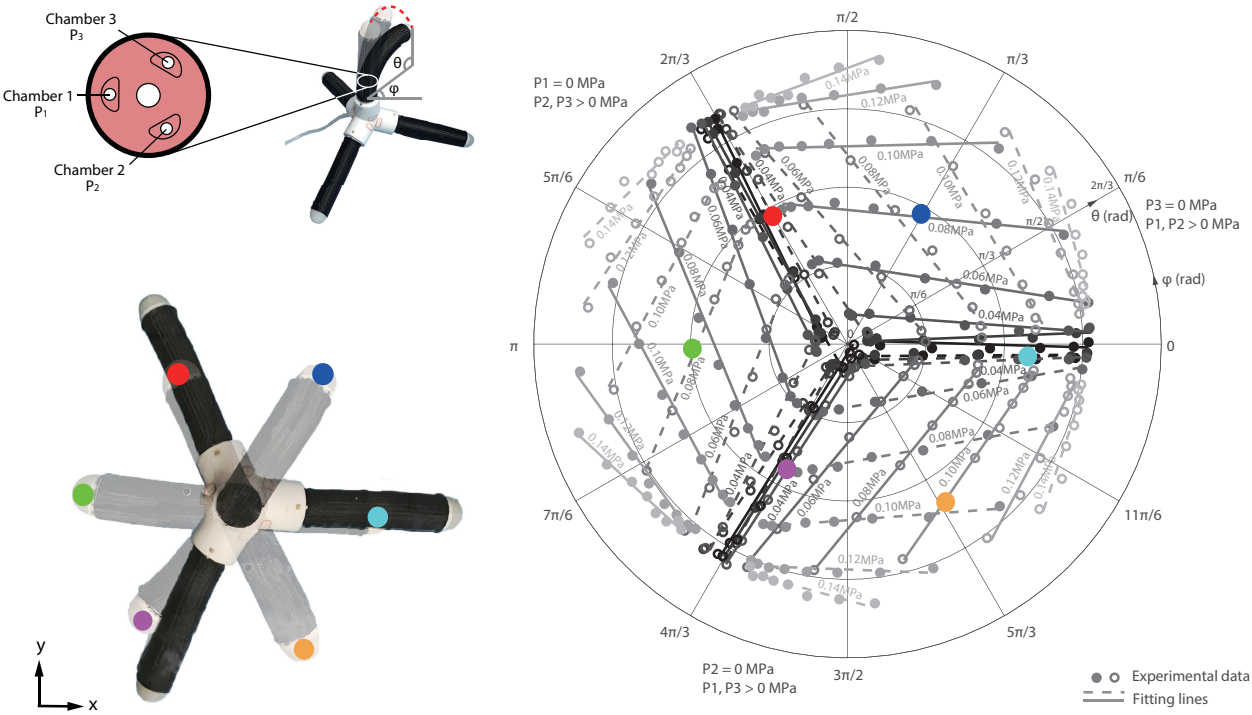


Fig. 2. **Fitting bending angles of the soft leg.** When filled with different air pressures, the soft actuators of the robot will have different degrees of bending deformation in the experiment. This picture shows the pressure deformation relationship.

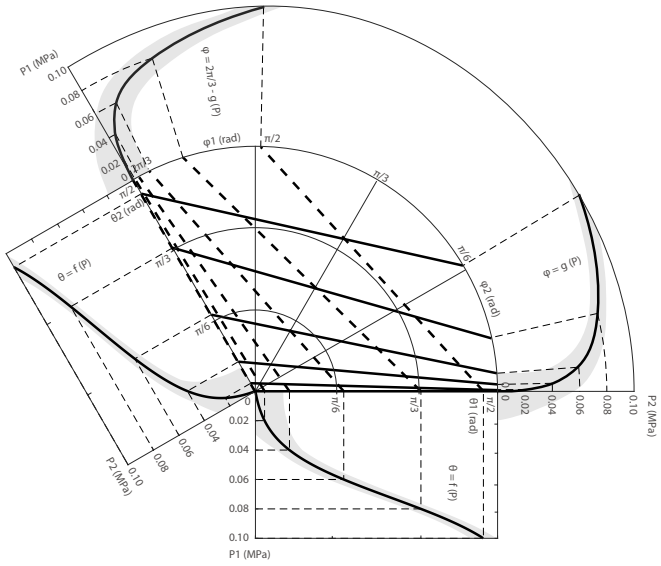


Fig. 3. **Relationship of the bending angle and inflating pressures.** We take one-third of them to study because of the symmetry.

B. Actuator model

1) Pressures-deformation relationship: For each soft actuator of the robot, the bending deformation occurs when the chambers are inflated with different pressures. The bending angle (θ) and direction (φ) are dependent on the inflating pressures of the three chambers (P_1, P_2, P_3), as shown in Fig.2. When chamber 1 and chamber 2 are inflated with arbitrary pressures, the soft actuator bends in direction range $[0, 2\pi/3]$. The bending direction ranges for inflation of chamber 2 and 3 and inflation of chamber 3 and chamber 1 are $[2\pi/3, 4\pi/3]$ and $[4\pi/3, 2\pi]$ respectively. We mark six points on the X-Y plane of the robot, and mark the corresponding six points in the polar diagram in Fig.2. From the polar plot of θ and φ (Fig.2), we find two clusters of isolines of inflating pressures intersect with each other in each bending direction range, and each intersection point indicates the bending angle and direction of the soft actuator under the corresponding inflating pressures of chambers. These data points in Fig.2 come from the robot experiment.

Considering the symmetry of the polar plot, we take one-third of it to be studied, as shown in Fig.3. On each solid isoline of pressure, P_1 varies while P_2 keeps constant. On each dashed isoline of pressure, P_2 varies while P_1 keeps constant. Each isoline is determined by its two endpoints on the radius and circle respectively. The coordinate θ_1 or θ_2 of the endpoints on the radius is related with the inflating pressure P_1 or P_2 by the function $\theta = f(P)$. The coordinate φ_1 or φ_2 of the endpoints on the circle is related to the inflating pressure P_1 or P_2 by the function $\varphi = g(P)$ or $\varphi = 2\pi/3 - g(P)$. The two functions are obtained by fitting experimental data shown in Fig.2, and the fitting curves and error bands are shown in Fig.3. The expressions of functions are shown in (1)-(4).

$$\theta_1 = f(P_1) = \left[1 - \frac{1 + e^{b_3 b_2}}{1 + e^{b_3 (b_2 - P_1)}}\right] b_1 \tag{1}$$

$$\theta_2 = f(P_2) \tag{2}$$

$$\varphi_2 = g(P_2) = \left[1 - \frac{1 + e^{c_3 c_2}}{1 + e^{c_3 (c_2 - P_2)}}\right] c_1 \tag{3}$$

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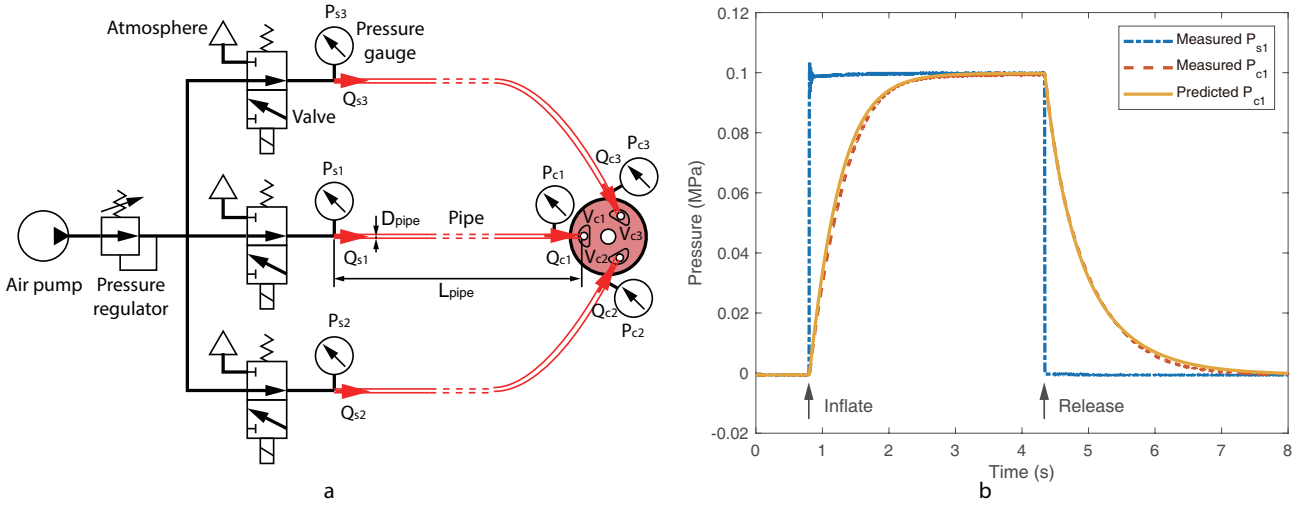


Fig. 4. **Pressure response through the pipe.** (a) The pneumatic system. (b) The delay between the supply pressure and the chamber pressure. This is caused by the resistance of pipes.

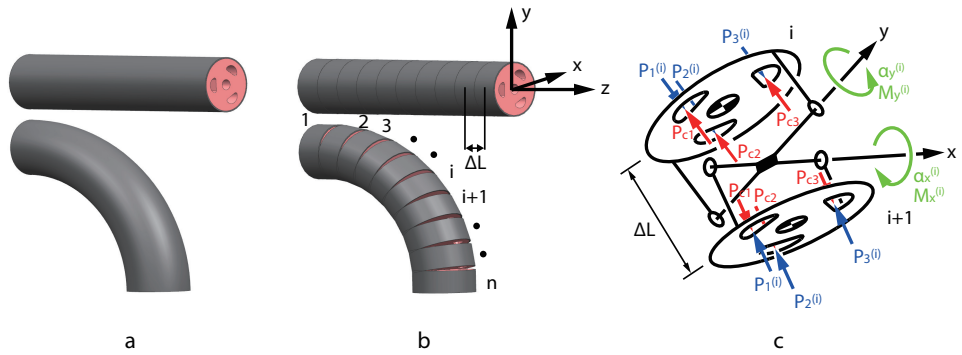


Fig. 5. **Discrete segments of the soft actuator.** (a) The continuous soft actuator. (b) The actuator is discretized into cylindrical rigid links with the same length. (c) Simplified mechanical model of the cylindrical rigid link.

$$\varphi_1 = \frac{2\pi}{3} - g(P_1) \tag{4}$$

where $b_1 = -0.02973$ rad, $b_2 = 0.07260$ MPa, $b_3 = 56.43$ MPa^{-1}, $c_1 = -0.002005$ rad, $c_2 = 0.1052$ MPa, $c_3 = 61.54$ MPa^{-1}.

The isolines of pressures P_1 and P_2 are expressed by (5) and (6) respectively.

$$\begin{aligned}&\theta\cos\left[\varphi - \tan^{-1}\frac{(2\pi/3)\cos\varphi_1 - \theta_1}{(2\pi/3)\sin\varphi_1}\right]\\&= \theta_1\cos\left[\tan^{-1}\frac{(2\pi/3)\cos\varphi_1 - \theta_1}{(2\pi/3)\sin\varphi_1}\right]\end{aligned} \tag{5}$$

$$\begin{aligned}&\theta\cos\left[2\pi/3 - \varphi - \tan^{-1}\frac{(2\pi/3)\cos(2\pi/3-\varphi_2) - \theta_2}{(2\pi/3)\sin(2\pi/3-\varphi_2)}\right]\\&= \theta_2\cos\left[\tan^{-1}\frac{(2\pi/3)\cos(2\pi/3-\varphi_2) - \theta_2}{(2\pi/3)\sin(2\pi/3-\varphi_2)}\right]\end{aligned} \tag{6}$$

Solving (5)-(6) yields the coordinates of the intersection point, i.e. θ and φ. It means that the soft actuator will bend by the angle θ along the direction φ under the inflating pressures P_1 and P_2. Replacing the angle θ with the curvature κ (in (7)), we get the expression of κ and φ with respect to P_1 and P_2, denoted by (8).

$$\kappa = \theta / L \tag{7}$$

$$(\kappa, \varphi) = h(P_1, P_2) \tag{8}$$

In the modeling, it is necessary to obtain the inflating pressures that just keep the bending deformation, which is the inverse problem of the aforementioned one. The equation for the direction range $[0, 2\pi/3]$, i.e. (9) can be extended into the range $[2\pi/3, 4\pi/3]$ and $[4\pi/3, 2\pi]$, that is (10) and (11) respectively.

$$(P_1, P_2) = h^{-1}(\kappa, \varphi), P_3 = 0, 0 \le \varphi < \frac{2\pi}{3} \tag{9}$$

$$(P_2, P_3) = h^{-1}(\kappa, \varphi - \frac{2\pi}{3}), P_1 = 0, \frac{2\pi}{3} \le \varphi < \frac{4\pi}{3} \tag{10}$$

$$(P_3, P_1) = h^{-1}(\kappa, \varphi - \frac{4\pi}{3}), P_2 = 0, \frac{4\pi}{3} \le \varphi < 2\pi \tag{11}$$

2) Pneumatic dynamics: In the pneumatic system (Fig.4a), the resistance of pipes induces a delay between the supply pressure P_{sj} and the chamber pressure P_{cj} (Fig.4b). According to Poiseuille law[42], the difference between P_{sj} and P_{cj} produces the flow rate through the pipe, as expressed by (12). Due to compressibility of air, the flow rate at the chamber

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TABLE I
SYMBOLS AND DESCRIPTIONS

Descriptions (the jth chamber, the ith cardan joint)	Symbols	Values
Length, diameter and mass of the pivot leg	$L_{\mathrm{p}}, D_{\mathrm{p}}, m_{\mathrm{p}}$	70 mm, 60 mm, 40 g
Length, diameter and mass of the soft leg	$L_{\mathrm{s}}, D_{\mathrm{s}}, m_{\mathrm{s}}$	170 mm, 42 mm, 240 g
Length, diameter and mass of the cap	$L_{\mathrm{c}}, D_{\mathrm{c}}, m_{\mathrm{c}}$	15 mm, 40 mm, 22 g
Length, diameter and mass of the toe	$L_{\mathrm{t}}, D_{\mathrm{t}}, m_{\mathrm{t}}$	20 mm, 40 mm, 19 g
Length and diameter of the pipe	$L_{\mathrm{pipe}}, D_{\mathrm{pipe}}$	1 mm, 3000 mm
The same length of the rigid links	ΔL	Variable (mm)
Major static moment of each air chamber cross section	S	
Bending angle of the soft leg	θ	Variable (rad)
Bending curvature of the soft leg	κ	Variable ($\mathrm{rad\,mm^{-1}}$)
The local bending curvature of the ith cardan joint	$\kappa^{(i)}$	Variable ($\mathrm{rad\,mm^{-1}}$)
Directional angle of the leg's bending	φ	Variable (rad)
The local bending direction of the ith cardan joint	$\varphi^{(i)}$	Variable (rad)
Rotation angles of the ith cardan joint	$\alpha_{\mathrm{x}}^{(i)}, \alpha_{\mathrm{y}}^{(i)}$	Variable (rad)
Number of discrete segments of the soft leg	n	10
Rotation moments of the ith cardan joint	$M_{\mathrm{x}}^{(i)}$, $M_{\mathrm{y}}^{(i)}$	Variable (N mm)
The local inflating pressure of the ith cardan joint of the jth chamber	$P_{\mathrm{j}}^{(i)}$	Variable (MPa)
Deformation pressures in the jth chambers of each soft leg	P_{j}	Variable (MPa)
Atmospheric pressure	P_{atm}	0.101 MPa
Air pressure supplied by the regulator of the jth chamber	P_{sj}	Variable (MPa)
Air pressure in the jth chamber	P_{cj}	Variable (MPa)
Air flow at the supply side of the jth chamber	Q_{sj}	Variable ($\mathrm{mm^3\,s^{-1}}$)
Air flow at the chamber side of the jth chamber	Q_{cj}	Variable ($\mathrm{mm^3\,s^{-1}}$)
Initial volume of each air chamber	V_{c0}	Variable (MPa)
Volume of the jth chambers of each soft leg	V_{j}	Variable ($\mathrm{mm^3}$)
The local variation of the ith cardan joint of the jth chamber volume	$\Delta V_{cj}^{(i)}$	Variable (mm)
Dynamic viscosity of air at 25 °C and 0.101 MPa	η	1.85×10^{-11} MPa · s
Frictional coefficients at three different positions : toe, soft actuator, pivot	$\mu_{\mathrm{t}}, \mu_{\mathrm{s}}, \mu_{\mathrm{p}}$	

side ($Q_{\mathrm{c}j}$) is different from that at the supply side ($Q_{\mathrm{s}j}$), and the flow rate is additionally related to the relative average pressure. The diameter and length of pipes (D_{pipe} and L_{pipe}) and viscosity of air (η) affect the resistance of pipes and thus also make influence on the flow rate.

$$Q_{\mathrm{c}j} = \frac{\pi D_{\mathrm{pipe}}^4}{128\eta L_{\mathrm{pipe}}} \frac{(P_{\mathrm{s}j} - P_{\mathrm{c}j})(P_{\mathrm{s}j} + P_{\mathrm{c}j} + 2P_{\mathrm{atm}})}{2(P_{\mathrm{s}j} + P_{\mathrm{atm}})} \tag{12}$$

where j means the jth chamber, $j = 1, 2, 3$.

By assuming the inflating and releasing processes are both isothermal, we can obtain the variation rate of the chamber pressure, as expressed by (13). It means both the airflow $Q_{\mathrm{c}j}$ and the volume variation $\dot{V}_{\mathrm{c}j}$ contribute to the pressure variation $\dot{P}_{\mathrm{c}j}$ of chambers. Substituting (12) into (13) yields the state transfer function of chamber pressure, i.e. (14).

$$\dot{P}_{\mathrm{c}j} = (Q_{\mathrm{c}j} - \dot{V}_{\mathrm{c}j})(P_{\mathrm{c}j} + P_{\mathrm{atm}})/V_{\mathrm{c}j} \tag{13}$$

$$\dot{P}_{\mathrm{c}j} = \Phi(P_{\mathrm{c}j}, P_{\mathrm{s}j}, V_{\mathrm{c}j}, \dot{V}_{\mathrm{c}j}) \tag{14}$$

where j means the jth chamber, $j = 1, 2, 3$.

Using (14) to predict the dynamic response of chamber pressure to supply pressure, we obtain the results shown in Fig.4b. The pneumatic system behaves almost like a first-order dynamic system, and the state transfer function above can predict the behavior of the pneumatic system with good accuracy.

3) Discrete link model: The bending curvature of the soft actuator can be inhomogeneous along its length especially when force load is exerted on it. It is difficult to solve the continuous distribution of curvature by using a spatially-continuous equation. Therefore, we consider utilizing the discrete link model to approximate the soft actuator, as shown in Fig.5.

The continuous soft actuator is discretized into n cylindrical rigid links with the same length ΔL (Fig.5b). Since the soft actuator is designed to bend without twisting, we assume Cardan joints between links. Components of the joint angle ($\alpha_{\mathrm{x}}^{(i)}$ and $\alpha_{\mathrm{y}}^{(i)}$) can be converted into the local bending curvature and direction ($\kappa^{(i)}$ and $\varphi^{(i)}$), as expressed by (15)-(18).

$$\kappa^{(i)} = \sqrt{(\alpha_{\mathrm{x}}^{(i)})^2 + (\alpha_{\mathrm{y}}^{(i)})^2}/\Delta L \tag{15}$$

$$\dot{\kappa}^{(i)} = (\alpha_{\mathrm{x}}^{(i)}\dot{\alpha}_{\mathrm{x}}^{(i)} + \alpha_{\mathrm{y}}^{(i)}\dot{\alpha}_{\mathrm{y}}^{(i)})/[\kappa^{(i)}(\Delta L)^2] \tag{16}$$

$$\varphi^{(i)} = \tan^{-1}(-\alpha_{\mathrm{x}}^{(i)}/\alpha_{\mathrm{y}}^{(i)}) \tag{17}$$

$$\dot{\varphi}^{(i)} = (\alpha_{\mathrm{x}}^{(i)}\dot{\alpha}_{\mathrm{y}}^{(i)} - \alpha_{\mathrm{y}}^{(i)}\dot{\alpha}_{\mathrm{x}}^{(i)})/(\kappa^{(i)}\Delta L)^2 \tag{18}$$

where i means the ith cardan joint, $i = 1, 2, 3, \ldots, n-1$.

Substituting (15) and (17) into (9)-(11) yields the local inflating pressure that just keeps the local bending deformation, i.e. $P_j^{(i)}$. The local bending deformation also induces a local variation of the chamber volume, i.e. $\Delta V_{cj}^{(i)}$ and $\Delta \dot{V}_{cj}^{(i)}$ in (19) and (20) respectively. The total volume of the chamber and its variation rate can be calculated by (21) and (22) respectively. Substituting (21)-(22) into (14), we obtain the variation rate of the chamber pressure $P_{\mathrm{c}j}$.

$$\Delta V_{\mathrm{c}j}^{(i)} = -S\Delta L\kappa^{(i)}\cos(\varphi^{(i)} - 2\pi j/3 - \pi/3) \tag{19}$$

$$\Delta \dot{V}_{\mathrm{c}j}^{(i)} = -S\Delta L[\dot{\kappa}^{(i)}\cos(\varphi^{(i)} - 2\pi j/3 - \pi/3)$$

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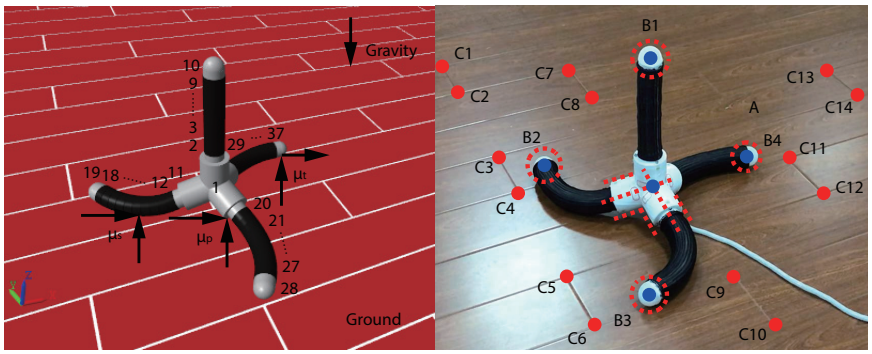


Fig. 6. **Simulation motion and robot motion capture.** (a) The motion of the multibody dynamics model of the robot in the simulation. (b) The motion of the robot in the experiment.

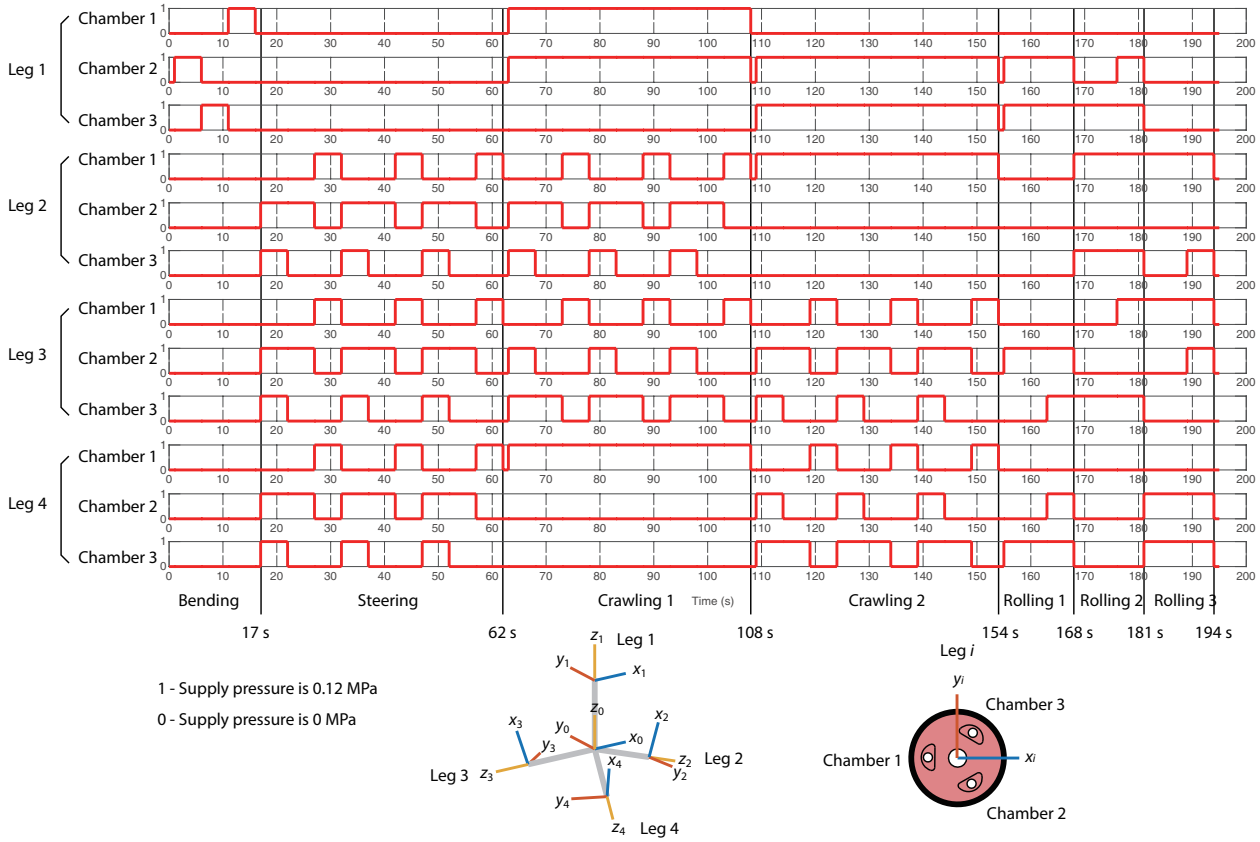


Fig. 7. **Actuation sequence.** The air pressure supply to the air chamber of the soft leg is supplied in sequence in the experiment.

$$- \kappa^{(i)}\dot{\varphi}^{(i)} \sin(\varphi^{(i)} - 2\pi j/3 - \pi/3)] \quad (20)$$

$$V_{cj} = V_{c0} + \sum_{i=1}^{n-1} \Delta V_{cj}^{(i)} \quad (21)$$

$$\dot{V}_{cj} = \sum_{i=1}^{n-1} \Delta \dot{V}_{cj}^{(i)} \quad (22)$$

where i means the ith cardan joint, $i = 1, 2, 3, \ldots, n-1$, and j means the jth chamber, $j = 1, 2, 3$.

Difference between the chamber pressure P_{cj} and the local inflating pressure $P_j^{(i)}$ produces the local bending moments or torque components of the Cardan joint, i.e. $M_x^{(i)}$ and $M_y^{(i)}$ in (23)-(24), which has been illustrated in Fig.5c.

$$M_x^{(i)} = \sum_{j=1}^{3} (P_{cj} - P_j^{(i)}) S \sin(2\pi j/3 + \pi/3) \quad (23)$$

$$M_y^{(i)} = -\sum_{j=1}^{3} (P_{cj} - P_j^{(i)}) S \cos(2\pi j/3 + \pi/3) \quad (24)$$

where i means the ith cardan joint, $i = 1, 2, 3, \ldots, n-1$, and j means the jth chamber, $j = 1, 2, 3$.

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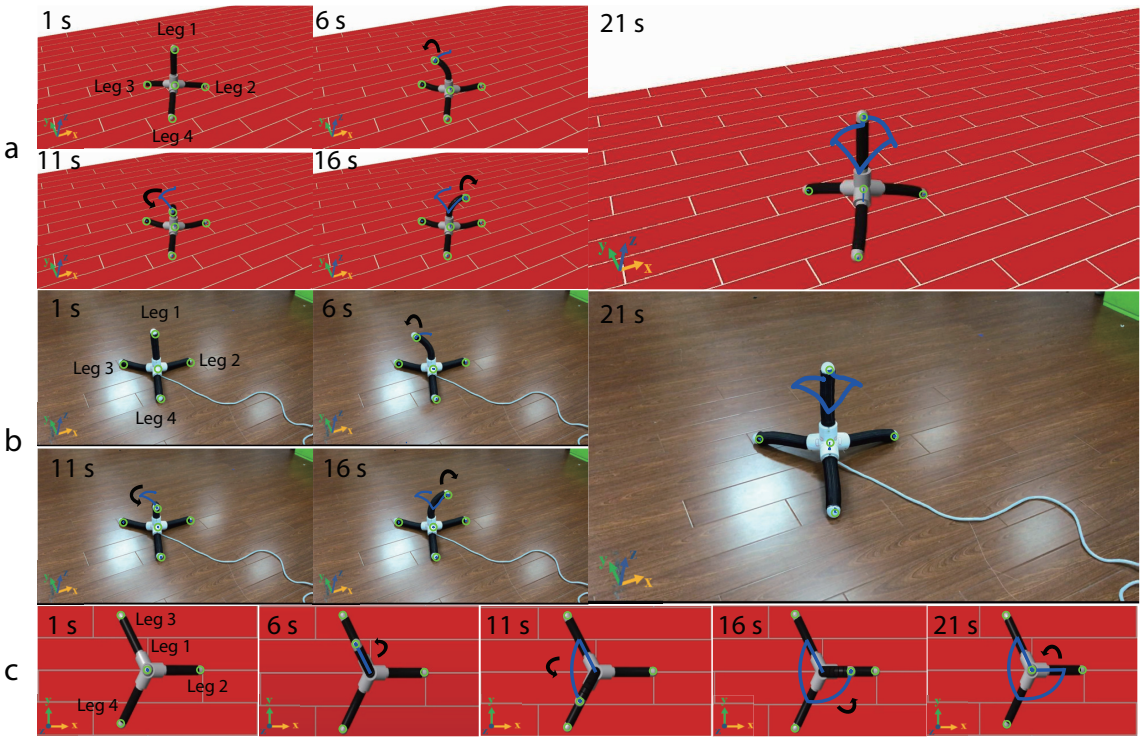


Fig. 8. **Bending trajectory.** (a) Bending motion in the simulation. (b) Bending motion in the experiment. The soft leg of the robot can bend in all directions. (c)Bending motion in X-Y plane.

C. Multibody dynamics

Based on the discrete link model of the soft actuator, the multibody dynamics model of the robot can be constructed in the simulation, as shown in Fig.6a. We divide the soft actuator into 10 rigid links in the discrete model and thus obtain 37 bodies in the dynamics model of the robot. Body 1 includes the pivot, tubes connector fittings, proximal caps and the first proximal link of the soft actuator. Body 10, 19, 28, 36 include the distal caps, toes and the first distal link of the soft actuator. Other bodies are identical discrete links of soft actuators. There are 36 Cardan joints in the multibody dynamics model. Contact can occur between the robot and ground at three different positions with different frictional coefficients, i.e. at the toe (μ_{t}), at the soft actuator (μ_{s}), and at the pivot (μ_{p}).

D. Motion capture

To capture the motion of the robot in the experiment, we locate the center point of the pivot by the intersection between central lines of two cylinders, i.e. point A shown in Fig.6b. The center points of toes are located by the center of a circle that encircles the toe i.e. points B1, B2, B3 and B4 in Fig.6b. The points C1, C2, ..., C14 on the ground are used to obtain position and view direction of the camera in experiments, and the animation of simulation is recorded with the same camera configure.

III. Simulation and Experiment

In simulation and experiment, the chambers of the soft legs are fed with the sequence of supply pressures as shown in the upper part of Fig.7. The lower part shows the convention for numbering the soft legs and chambers. Considering the strength of soft actuators, we limit the maximum inflating pressure to $0.12\,\mathrm{MPa}$. In the dynamic simulation, we use Coulomb friction model to simulate the friction in real environment. From time $0\,\mathrm{s}$ to $200\,\mathrm{s}$, the robot first performs bending deformation of a soft leg and then performs the steering, crawling and rolling gaits of locomotion sequentially. By comparing the two trajectories, the consistency of simulation and experiment is ensured. The motion of the robot is captured by the aforementioned vision method, and the results are shown in Fig.8-11.

A. Bending motion

The soft leg of the robot can perform omnidirectional bending motion, as shown in Fig.8. The tip trajectory of the soft leg in simulation is consistent with that in the experiment. During the bending motion, the three chambers of the soft leg are inflated and released in turn, as shown in Fig.7. The bending motion of soft legs would allow the robot to obtain the panorama of the environment if the soft legs are integrated with vision sensors, or actively touch and sense environmental objects if the soft legs are integrated with tactile sensors.

B. Steering motion

During steering locomotion, the three soft legs contacting with the ground (i.e. Legs 2, 3 and 4) have their chambers to be inflated and released with the same sequence (Fig.7), and thus exhibit the same bending motion (Fig.9). The three anchoring legs first bend downward to elevate the robot body off the

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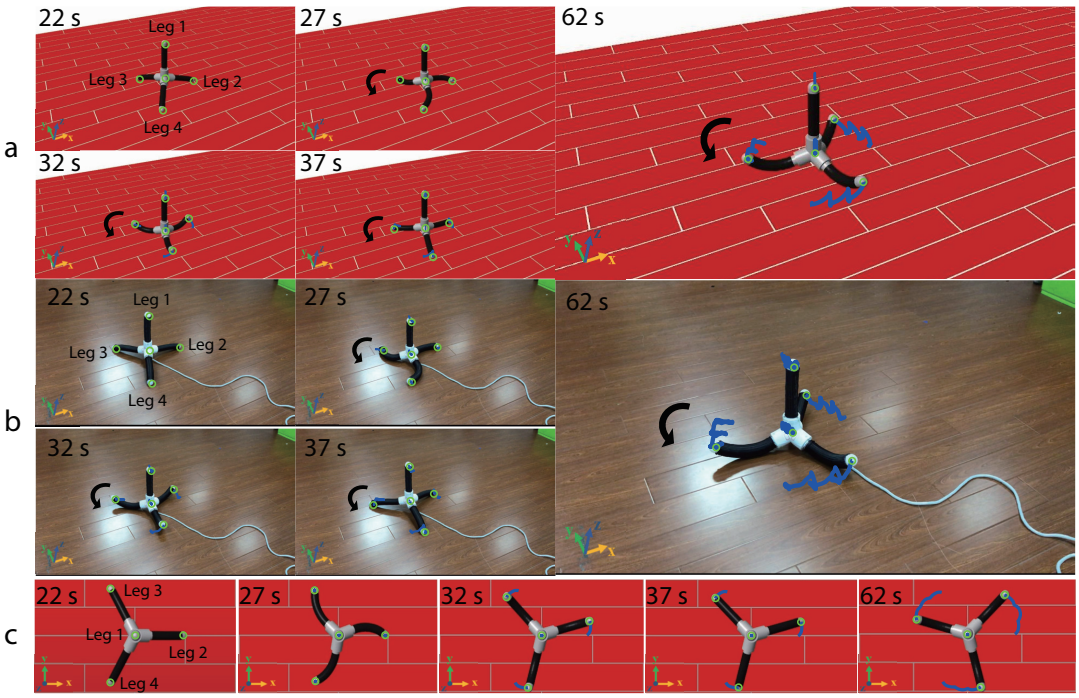


Fig. 9. **Steering trajectory.** (a) Steering motion in the simulation. (b) Steering motion in the experiment. (c)Steering motion in X-Y plane.

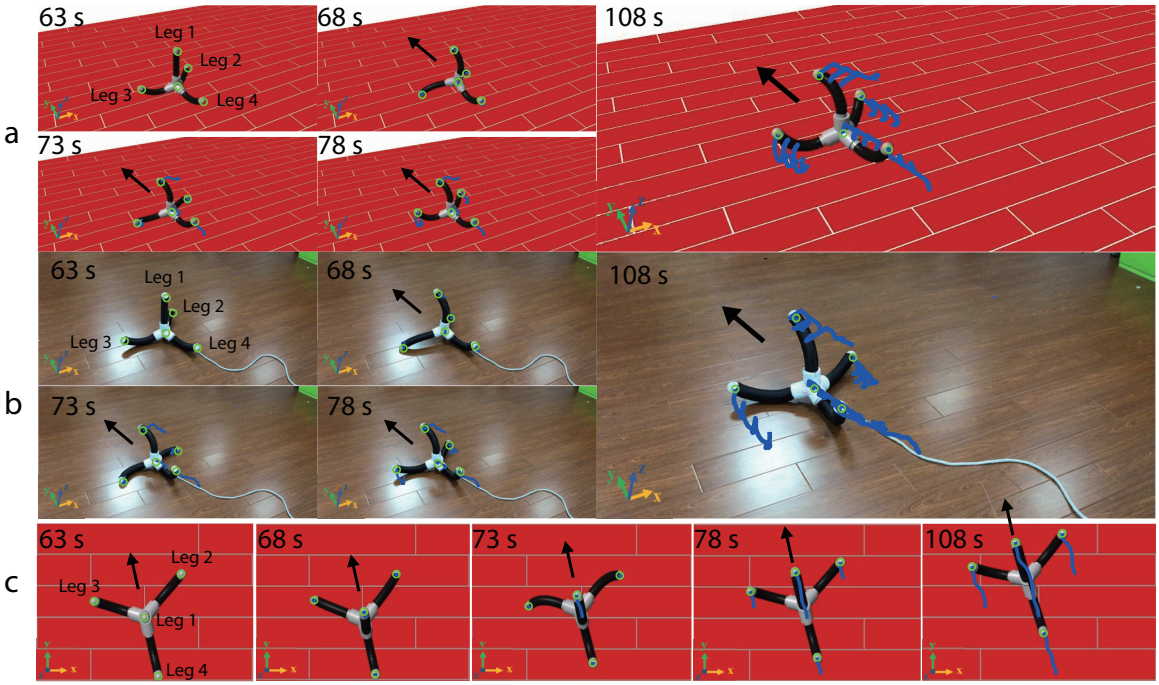


Fig. 10. **Crawling trajectory.** (a) Crawling motion in the simulation. (b) Crawling motion in the experiment. The robot can crawl in three directions. (c)Crawling motion in X-Y plane.

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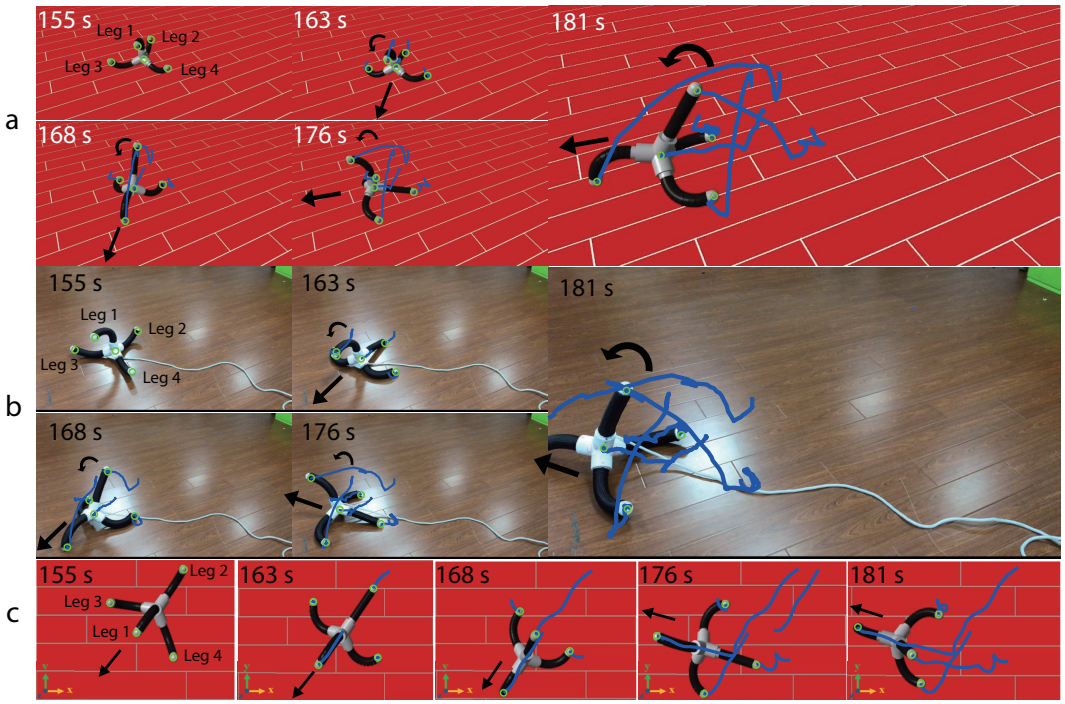


Fig. 11. **Rolling trajectory.** (a) Rolling motion in the simulation. (b) Rolling motion in the experiment. The robot can roll in three directions. (c)Rolling motion in X-Y plane.

ground (at 22 s), then bend clockwise to rotate the robot body counterclockwise (at 27 s), next bending upward to put down the robot body on the ground (at 32 s), and the last bend down again to repeat the steering process (at 37 s). After 2.5 periods of the steering process, the robot has turned counterclockwise by an angle of 45° (at 62 s). The steering speed is $1.1\,^{\circ}\,\mathrm{s}^{-1}$. Clockwise steering can be achieved by making the three soft legs bend counterclockwise at 27 s. The steering locomotion allows the robot to change its pose and moving direction while keeping the current position, which is useful for search and rescue in limited space.

C. Crawling motion

The proposed tetrahedral soft-legged robot can perform crawling locomotion along three directions, and Fig.10 shows one of them. During crawling locomotion, Leg 1 keeps bending forward for shifting the mass center of the robot toward Leg 2 and Leg3. At the same time, Leg 4 keeps bending upward for elevating its toe-off ground and thus reducing the rear frictional force. Leg 2 and Leg 3 first bend upward for putting down the robot body (at 63 s), then bend downward for elevating the robot body (at 68 s), next bend backward for pushing the robot forward (at 73 s), and last bend upward again for repeating the crawling process (at 78 s). After 3 periods of crawling locomotion, the robot moves forward by a distance of 167 mm (at 108 s). The crawling speed is $3.7\,\mathrm{mm\,s}^{-1}$. Crawling locomotion along the other two directions can be achieved by the bending motion of Legs 2 and 4 and Legs 3 and 4 respectively. The proposed tetrahedral soft-legged always has three contact points with the ground during crawling locomotion, and thus exhibits better stability than the previous inchworm-inspired crawling robot.

D. Rolling motion

The proposed tetrahedral soft-legged robot can perform rolling locomotion along three directions, and Fig.11 shows two of them. During the first rolling locomotion, Leg 1 bend forward for shifting the mass center of the robot forward (at 155 s), and then Leg 3 and Leg 4 bend backward for shifting the two front contact points with ground backward (at 163 s). When the mass center surpasses the front contact points, the robot will roll around the front contact points (at 168 s), i.e. the rolling locomotion. The second rolling locomotion (from 168 s to 181 s) has the same working principle, except with a different rolling direction. In each rolling movement, the robot rolls by an angle of 71° and a distance of 268 mm. The rolling speed is $5.5\,^{\circ}\,\mathrm{s}^{-1}$ and $20.6\,\mathrm{mm\,s}^{-1}$.

IV. Conclusions

The tetrahedral soft-legged robot for multi-mode locomotion is proposed in this paper. The tetrahedral structure design allows the robot to stand and move stably with just three legs, and the four soft actuators can provide a shorter actuation sequence and faster speed for the robot. The pressure-deformation relationship, pneumatic dynamics and multi-body dynamics of the robot are analyzed with an soft actuator. Through simulation and experiment, the feasibility of the proposed robot and the modeling method are validated. The soft legs of the robot are actuated by different inflation sequences to perform omnidirectional bending motions and realize multi gait locomotion, including steering, crawling and rolling locomotion. The crawling speed of the robot can reach $3.7\,\mathrm{mm\,s}^{-1}$, and it can achieve a steering motion of $1.1\,^{\circ}\,\mathrm{s}^{-1}$, and complete a rolling motion of $5.5\,^{\circ}\,\mathrm{s}^{-1}$ and $20.6\,\mathrm{mm\,s}^{-1}$.

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The main contributions of this work include: (1) proposing a novel tetrahedral structure and the novel soft actuator with three air chambers; (2) proposing a discrete link model to calculate the dynamics of the soft actuator; (3) developing a control scheme and achieving effective control of the multi gait locomotion of the proposed tetrahedral soft-legged robot.

The proposed robots also has some gaps. And in future work, there are several aspects of this research to be improved. First, to improve the adaptability of the robot in complex environments, we will explore more gaits for the robot and develop a gait pattern modulator to achieve smooth switching of different gaits. Second, to improve the perception and control performance of the robot, we will integrate more sensors into the robot to increase the environment perception and motion perception capabilities and achieve closed-loop control for the robot. Third, to increase the application scenarios of robots, we will design it as untethered. So that we need to solve the problem of power supply for the robot. Forth, to increase the speed of the robot, we will use different types of trachea to improve inflation and deflation speed. The proposed tetrahedral soft-legged robot has potential applications in search and rescue.

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